ANSWERS

1.

(a) **(i)** $pV = nRT$

$V = (8.31 \times 300)/(1.02 \times 10^5)$ C1

$= 0.0244\ m^3$ (if uses Celsius, then 0/2) A1

(ii) volume occupied by one atom $= 0.0244 / (6.02 \times 10^{23}) = 4.06 \times 10^{-26}\ m^3$ M1

separation $\approx \sqrt[3]{(4.06 \times 10^{-26})}$ A1

$= 3.44 \times 10^{-9}$ m A0

(b) **(i)** $F = GMm / r^2$ C1

$= (6.67 \times 10^{-11} \times \{4 \times 1.66 \times 10^{-27}\}^2) / (3.44 \times 10^{-9})^2$ C1

$= 2.49 \times 10^{-46}$ N A1

(ii) ratio $= (4 \times 1.66 \times 10^{-27} \times 9.8) / 2.49 \times 10^{-46}$ C1

$= 2.6 \times 10^{20}$ A1

(c) assumption that forces between atoms are negligible B1

comment e.g. ratio shows gravitational force to be very small

e.g. force is very much less than weight

e.g. if there are forces, they are not gravitational B1

2.

(a) molecule(s) rebound from wall of vessel / hits walls B1

change in momentum gives rise to impulse / force B1

either (many impulses) averaged to give constant force / pressure

or the molecules are in random motion B1

(b) **(i)** $p = \frac{1}{3}\rho\langle c^2\rangle$ C1

$1.02 \times 10^5 = \frac{1}{3} \times 0.900 \times \langle c^2\rangle$

$\langle c^2\rangle = 3.4 \times 10^5$ C1

$c_{RMS} = 580\ m\ s^{-1}$ A1

(ii) *either* $\langle c^2\rangle \propto T$ *or* $\langle c^2\rangle = 2 \times 3.4 \times 10^5$ C1

$c_{RMS} = 830\ m\ s^{-1}$ (allow 820) A1

(c) c_{RMS} depends on temperature (alone) B1

so no effect B1

3.

(a) *either* $pV = NkT$ *or* $pV = nRT$ <u>*and*</u> $n = N / N_A$ C1

clear correct substitution e.g.

$2.5 \times 10^5 \times 4.5 \times 10^3 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 290$ M1

$N = 2.8 \times 10^{23}$ A0

(*allow 1 mark for calculation of n = 0.467 mol*)

(b) (i) volume $= (1.2 \times 10^{-10})^3 \times 2.8 \times 10^{23}$ *or* $\frac{4}{3}\pi r^3 \times 2.8 \times 10^{23}$ C1

$= 4.8 \times 10^{-7}\ m^3$ $\quad$ $2.53 \times 10^{-7}\ m^3$ A1

(ii) *either* $4.5 \times 10^3\ cm^3 >> 0.48\ cm^3$ *or* ratio of volumes is about 10^{-4} B1

justified because volume of molecules is negligible B1

4.

(a) (i) 27.2 + 273.15 or 27.2 + 273.2 C1

300.4 K A1

(ii) 11.6 K A1

(b) (i) ($<c^2>$ is the) mean / average square speed B1

(ii) $\rho = Nm/V$ with <u>N explained</u> B1

so, $pV = 1/3\ Nm<c^2>$ B1

and $pV = NkT$ with <u>k explained</u> B1

<u>so mean kinetic energy / $<E_K> = ½m<c^2> = 3/2\ kT$</u> B1

(c) (i) $pV = nRT$

$2.1 \times 10^7 \times 7.8 \times 10^{-3} = n \times 8.3 \times 290$ C1

$n = 68$ mol A1

(ii) mean kinetic energy $= 3/2\ kT$

$= 3/2 \times 1.38 \times 10^{-23} \times 290$ C1

$= 6.0 \times 10^{-21}$ J A1

(iii) realisation that total internal energy is the total kinetic energy C1

energy $= 6.0 \times 10^{-21} \times 68 \times 6.02 \times 10^{23}$ C1

$= 2.46 \times 10^5$ J A1

5.

(a) atoms / molecules / particles behave as elastic (identical) spheres (1)
volume of atoms / molecules negligible compared to volume of containing vessel (1)
time of collision negligible to time between collisions (1)
no forces of attraction or repulsion between atoms / molecules (1)
atoms / molecules / particles are in (continuous) random motion (1)
(*any four, 1 each*) B4

(b) $pV = \frac{1}{3}Nm<c^2>$ and $pV = nRT$ *or* $pV = NkT$ B1
$\frac{1}{3}Nm<c^2> = nRT$ or $= NkT$ and $<E_K> = \frac{1}{2}m<c^2>$ B1
$n = N/N_A$ *or* $k = R/N_A$ B1
$<E_K> = \dfrac{3}{2} \times R/N_A \times T$ A0

(c) **(i)** reaction represents *either* build-up of nucleus from light nuclei
or build-up of heavy nucleus from nuclei M1
so fusion reaction A1

(ii) proton and deuterium nucleus will have equal kinetic energies B1
$1.2 \times 10^{-14} = \frac{3}{2} \times 8.31 / (6.02 \times 10^{23}) \times T$ C1
$T = 5.8 \times 10^8$ K A1
(*use of* $E = 2.4 \times 10^{-14}$ *giving* 1.16×10^9 *K scores 1 mark*)

(iii) *either* inter-molecular / atomic / nuclear forces exist
or proton and deuterium nucleus are positively charged / repel B1

6.

(a) number of atoms of carbon-12 M1
in 0.012 kg of carbon-12 A1

(b) $pV = NkT$ or $pV = nRT$ C1
substitutes temperature as 298 K C1
either $1.1 \times 10^5 \times 6.5 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times 298$
or $1.1 \times 10^5 \times 6.5 \times 10^{-2} = n \times 8.31 \times 298$ and $n = N / 6.02 \times 10^{23}$ C1
$N = 1.7 \times 10^{24}$ A1

7.

(a) amount of substance — M1

containing same number of particles as in 0.012 kg of carbon-12 — A1

(b) $pV = nRT$ — C1

amount = $(2.3 \times 10^5 \times 3.1 \times 10^{-3}) / (8.31 \times 290)$

$+ (2.3 \times 10^5 \times 4.6 \times 10^{-3}) / (8.31 \times 303)$ — C1

= 0.296 + 0.420 — C1

= 0.716 mol — A1

(give full credit for starting equation $pV = NkT$ and $N = nN_A$)

8.

(a) e.g. moving in random (rapid) motion of <u>molecules/atoms/particles</u>

no intermolecular forces of attraction/repulsion

volume of <u>molecules/atoms/particles</u> negligible <u>compared</u> to volume of container

time of collision negligible to time between collisions

(1 each, max 2) — B2

(b) **(i)** **1.** number of (gas) <u>molecules</u> — B1

2. mean square speed/velocity (of gas molecules) — B1

(ii) either $pV = NkT$ *or* $pV = nRT$ and links n and k

and $\langle E_K \rangle = \frac{1}{2}m\langle c^2 \rangle$ — M1

clear algebra leading to $\langle E_K \rangle = \dfrac{3}{2}kT$ — A1

(c) **(i)** sum of potential energy and kinetic energy of <u>molecules/atoms/particles</u> — M1

reference to random (distribution) — A1

(ii) no intermolecular forces so no potential energy — B1

(change in) internal energy is (change in) kinetic energy and this is proportional to (change in) T — B1

9.

(a) **(i)** weight = GMm/r^2 C1

= $(6.67 \times 10^{-11} \times 6.42 \times 10^{23} \times 1.40)/(½ \times 6.79 \times 10^{6})^2$ C1

= 5.20 N A1

(ii) potential energy = $-GMm/r$ C1

= $-(6.67 \times 10^{-11} \times 6.42 \times 10^{23} \times 1.40)/(½ \times 6.79 \times 10^{6})$ M1

= -1.77×10^{7} J A0

(b) *either* $½mv^2 = 1.77 \times 10^7$ C1

$v^2 = (1.77 \times 10^7 \times 2)/1.40$ C1

$v = 5.03 \times 10^3\ \mathrm{m\,s^{-1}}$ A1

or $½mv^2 = GMm/r$ (C1)

$v^2 = (2 \times 6.67 \times 10^{-11} \times 6.42 \times 10^{23})/(6.79 \times 10^6/2)$ (C1)

$v = 5.02 \times 10^3\ \mathrm{m\,s^{-1}}$ (A1)

(c) **(i)** $½ \times 2 \times 1.66 \times 10^{-27} \times (5.03 \times 10^3)^2 = \frac{3}{2} \times 1.38 \times 10^{-23} \times T$ C1

T = 2030 K A1

(ii) *either* because there is a range of speeds M1

some molecules have a higher speed A1

or some escape from point above planet surface (M1)

so initial potential energy is higher (A1)

10.

(a) **(i)** *either* random motion

or constant velocity until hits wall/other molecule B1

(ii) (total) volume of molecules is negligible M1

compared to volume of containing vessel A1

or

radius/diameter of a molecule is negligible (M

compared to the average intermolecular distance (A

(b) *either* molecule has component of velocity in three directions

or $c^2 = c_X^2 + c_Y^2 + c_Z^2$ M1

random motion and averaging, so $\langle c_X^2 \rangle = \langle c_Y^2 \rangle = \langle c_Z^2 \rangle$ M1

$\langle c^2 \rangle = 3\langle c_X^2 \rangle$ A1

so, $pV = ⅓Nm\langle c^2 \rangle$ A0

(c) $\langle c^2 \rangle \propto T$ or $c_{rms} \propto \sqrt{T}$ C1

temperatures are 300 K and 373 K C1

$c_{rms} = 580\ \mathrm{m\,s^{-1}}$ A1

11.

(a) (i) number of molecules B1

(ii) mean square speed B1

(b) (i) 1. $pV = nRT$ C1
$n = (6.1 \times 10^5 \times 2.1 \times 10^4 \times 10^{-6}) / (8.31 \times 285)$ C1
$n = 5.4$ mol A1

2. *either* $N = nN_A$
$= 5.4 \times 6.02 \times 10^{23}$ C1
$= 3.26 \times 10^{24}$ A1
or
$pV = NkT$
$N = (6.1 \times 10^5 \times 2.1 \times 10^4 \times 10^{-6}) / (1.38 \times 10^{-23} \times 285)$ (C1)
$N = 3.26 \times 10^{24}$ (A1)

(ii) *either* $6.1 \times 10^5 \times 2.1 \times 10^{-2} = {}^1/_3 \times 3.25 \times 10^{24} \times 4 \times 1.66 \times 10^{-27} \times \langle c^2 \rangle$ C1
$\langle c^2 \rangle = 1.78 \times 10^6$ C1
$c_{RMS} = 1.33 \times 10^3\ \mathrm{m\,s^{-1}}$ A1
or
${}^1/_2 \times 4 \times 1.66 \times 10^{-27} \times \langle c^2 \rangle = {}^3/_2 \times 1.38 \times 10^{-23} \times 285$ (C1)
$\langle c^2 \rangle = 1.78 \times 10^6$ (C1)
$c_{RMS} = 1.33 \times 10^3\ \mathrm{m\,s^{-1}}$ (A1)

12.

(a) obeys the equation pV = constant $\times$ T or $pV = nRT$ M1
p, V and T explained A1
at all values of p, V and T/fixed mass/n is constant A1

(b) (i) $3.4 \times 10^5 \times 2.5 \times 10^3 \times 10^{-6} = n \times 8.31 \times 300$ M1
$n = 0.34$ mol A0

(ii) for total mass/amount of gas
$3.9 \times 10^5 \times (2.5 + 1.6) \times 10^3 \times 10^{-6} = (0.34 + 0.20) \times 8.31 \times T$ C1
$T = 360$ K A1

(c) when tap opened
gas passed (from cylinder B) to cylinder A B1
work done <u>on</u> gas in cylinder A (and no heating) M1
so internal energy and hence temperature increase A1

13.

(a) (i) 1. $pV = nRT$

$1.80 \times 10^{-3} \times 2.60 \times 105 = n \times 8.31 \times 297$ C1

$n = 0.19$ mol A1

2. $\Delta q = mc\Delta T$

$95.0 = 0.190 \times 12.5 \times \Delta T$ B1

$\Delta T = 40$ K A1

(allow 2 marks for correct answer with clear logic shown)

(ii) p/T = constant

$(2.6 \times 10^5) / 297 = p / (297 + 40)$ M1

$p = 2.95 \times 10^5$ Pa A0

(b) change in internal energy is 120 J / 25 J B1

internal energy decreases / $\underline{\Delta}U$ is negative / kinetic energy of molecules decreases M1

so temperature lower A1

14.

(a) (i) N: (total) number of <u>molecules</u> B1

(ii) $<c^2>$: mean square speed/velocity B1

(b) $pV = \frac{1}{3}Nm<c^2> = NkT$

(mean) kinetic energy = $\frac{1}{2} m<c^2>$ C1

algebra clear leading to $\frac{1}{2} m<c^2> = (3/2)kT$ A1

(c) (i) *either* energy required = $(3/2) \times 1.38 \times 10^{-23} \times 1.0 \times 6.02 \times 10^{23}$ C1

= 12.5 J (*12J if 2 s.f.*) A1

or energy = $(3/2) \times 8.31 \times 1.0$ (C1)

= 12.5 J (A1)

(ii) energy is needed to push back atmosphere/do work against atmosphere M1

so total energy required is greater A1

15.

(a) the number of atoms — M1
in 12 g of carbon-12 — A1

(b) **(i)** amount = 3.2/40
= 0.080 mol — A1

(ii) $pV = nRT$
$p \times 210 \times 10^{-6} = 0.080 \times 8.31 \times 310$ — C1
$p = 9.8 \times 10^{5}$ Pa — A1
(do not credit if T in °C not K)

(iii) *either* $pV = 1/3 \times Nm <c^2>$
$N = 0.080 \times 6.02 \times 10^{23}$ (= 4.82×10^{22})
and $m = 40 \times 1.66 \times 10^{-27}$ (= 6.64×10^{-26}) — C1
$9.8 \times 10^{5} \times 210 \times 10^{-6} = 1/3 \times 4.82 \times 10^{22} \times 6.64 \times 10^{-26} \times <c^2>$ — C1
$<c^2> = 1.93 \times 10^{5}$
$c_{RMS} = 440$ m s^{-1} — A1

or $Nm = 3.2 \times 10^{-3}$ — (C1)
$9.8 \times 10^{5} \times 210 \times 10^{-6} = 1/3 \times 3.2 \times 10^{-3} \times <c^2>$ — (C1)
$<c^2> = 1.93 \times 10^{5}$
$c_{RMS} = 440$ m s^{-1} — (A1)

or $1/2\ m<c^2> = 3/2\ kT$ — (C1)
$1/2 \times 40 \times 1.66 \times 10^{-27} <c^2> = 3/2 \times 1.38 \times 10^{-23} \times 310$ — (C1)
$<c^2> = 1.93 \times 10^{5}$
$c_{RMS} = 440$ m s^{-1} — (A1)

(*if T in °C not K award max 1/3, unless already penalised in* ***(b)(ii)***)

16.

(a) use of kelvin temperatures — B1
both values of (V/T) correct (11.87), V/T is constant so pressure is constant — M1

(*allow use of n =1. Do not allow other values of n.)*

(b) **(i)** work done $= p\Delta V$
$= 4.2 \times 10^{5} \times (3.87 - 3.49) \times 10^{3} \times 10^{-6}$ — C1
$= 160$ J — A1

(do not allow use of V instead of ΔV)

(ii) increase/change in internal energy = heating of system
+ work done on system — C1
= 565 – 160
= 405 J — A1

(c) internal energy = sum of kinetic energy and potential energy / $E_K + E_P$ B1
no intermolecular forces M1
no potential energy (so $\Delta U = \Delta E_K$) A1

17.

(a) obeys the equation pV/T = constant B1
(accept $pV = nRT$)

(b) **(i)** $pV = nRT$ C1
$5.0 \times 10^7 \times 3.0 \times 10^{-4} = n \times 8.31 \times 296$ giving $n = 6.1$ mol A1

(ii) pressure $\propto$ amount of substance
loss = 0.40 / 100 × 6.1 mol = 0.0244 mol C1
= $0.0244 \times 6.02 \times 10^{23}$ (atoms) C1
= 1.47×10^{22} atoms C1

rate = $(1.47 \times 10^{22}) / (35 \times 24 \times 60 \times 60)$
= $4.9 \times 10^{15}\ s^{-1}$ A1

18.

(a) initially, $pV/T = (2.40 \times 10^5 \times 5.00 \times 10^{-4}) / 288 = 0.417$ M1
finally, $pV/T = (2.40 \times 10^5 \times 14.5 \times 10^{-4}) / 835 = 0.417$ M1
ideal gas because pV/T is constant A1
(*allow 2 marks for two determinations of V / T and then 1 mark for V / T <u>and</u> p constant, so ideal*)

(b) **(i)** work done = $p\Delta V$
= $2.40 \times 10^5 \times (14.5 - 5.00) \times 10^{-4}$ C1
= 228 J (*ignore sign, not 2 s.f.*) A1

(ii) $\Delta U = q + w$ = 569 – 228
= 341 J M1
increase A1

19.

(a) mean speed = $1.44 \times 10^3\ m\,s^{-1}$ A1

(b) evidence of summing of individual squared speeds C1
mean square speed = $2.09 \times 10^6\ m^2\,s^{-2}$ A1

(c) root-mean-square speed = $1.45 \times 10^3\ m\,s^{-1}$ A1
*(allow ECF from **(b)** but only if arithmetic error)*

20.

(a) obeys the equation $pV = nRT$ *or* pV/T = constant — M1

all symbols explained; T in kelvin/thermodynamic temperature — A1

(b) (i) temperature rise = 48 K — A1

(ii) $<c^2> \propto T$ or equivalent — C1

$<c^2> = (353/305) \times 1.9 \times 10^6$ — C1

$c_{\text{r.m.s.}} = 1480\,\text{m s}^{-1}$ — A1

21.

(a) total volume of molecules negligible compared to that of containing vessel

no intermolecular forces

molecules in random motion

time of collision small compared with the time between collisions

large number of molecules

any two — B2

(b) in a real gas there is a range of velocities *or* must take the average of v^2 — B1

(c) (i) *either* $p = \frac{1}{3}\rho<c^2>$

or $1.0 \times 10^5 = \frac{1}{3} \times 1.2 \times <c^2>$ — C1

$<c^2> = 2.5 \times 10^5$ — C1

$c_{\text{r.m.s.}} = 500\,\text{m s}^{-1}$ — A1

(ii) $T \propto <c^2>$ — C1

$<c^2> = 2.5 \times 10^5 \times 480/300$

$= 4.0 \times 10^5\,\text{m}^2\,\text{s}^{-2}$ (*allow ECF from* ***(c)(i)***) — A1

22.

(a) (i) sum of kinetic and potential energy of atoms/molecules — M1

reference to random (distribution) — A1

(ii) no forces (of attraction or repulsion) between molecules — B1

(b) $pV = NkT$ *or* $pV = nRT$ *and* $R = kN_A$, $n = N/N_A$ — B1

$\frac{1}{3} Nm<c^2> = NkT$ *or* $\frac{1}{3} m<c^2> = kT$ — B1

$<E_K> = \frac{1}{2} m<c^2>$ <u>so</u> $<E_K> = \frac{3}{2} kT$ — B1

(c) **(i)** $<E_K> = {}^3/_2 \times 1.38 \times 10^{-23} \times (273 + 12)$ C1
$= 5.9\ (5.90) \times 10^{-21}$ J A1

(*use of* T = 12 K *not* T = 285 K scores 0/2)

(ii) number $= (17/32) \times 6.02 \times 10^{23}$ C1
$= 3.2\ (3.20) \times 10^{23}$ A1

(iii) internal energy $= 5.9 \times 10^{-21} \times 3.2 \times 10^{23}$
$= 1900\ (1890)$ J A1

23.

(a) e.g. time of collisions negligible compared to time between collisions

no intermolecular forces (except during collisions)

random motion (of molecules)

large numbers of molecules

(total) volume of molecules negligible compared to volume of containing vessel
or
average/mean separation large compared with size of molecules

any two B2

(b) **(i)** mass $= 4.0 / (6.02 \times 10^{23}) = 6.6 \times 10^{-24}$ g
or
mass $= 4.0 \times 1.66 \times 10^{-27} \times 10^{3} = 6.6 \times 10^{-24}$ g B1

(ii) $\frac{3}{2}kT = \frac{1}{2}m<c^2>$ C1

$\frac{3}{2} \times 1.38 \times 10^{-23} \times 300 = \frac{1}{2} \times 6.6 \times 10^{-27} \times <c^2>$

$<c^2> = 1.88 \times 10^{6}\ (\mathrm{m^2\,s^{-2}})$ C1

r.m.s. speed $= 1.4 \times 10^{3}\ \mathrm{m\,s^{-1}}$ A1

24.

(a) **(i)** number of <u>atoms/nuclei</u> in 12 g of carbon-12 B1

(ii) amount of substance M1

containing N_A (or 6.02×10^{23}) particles/molecules/atoms
or
which contains the same number of particles/atoms/molecules as there are atoms in 12 g of carbon-12 A1

(b) $pV = nRT$

$2.0 \times 10^7 \times 1.8 \times 10^4 \times 10^{-6} = n \times 8.31 \times 290$, so $n = 149$ mol or 150 mol A1

(c) **(i)** V and T constant and so pressure reduced by 5.0%
pressure = $0.95 \times 2.0 \times 10^7$ C1

or

calculation of new n (= 142.5 mol) and correct substitution into $pV = nRT$ (C1)

pressure = 1.9×10^7 Pa A1

(ii) loss is 5 / 100 × 150 mol = 7.5 mol
or
$\Delta N = 4.52 \times 10^{24}$ C1

$t = (7.5 \times 6.02 \times 10^{23}) / 1.5 \times 10^{19}$
or
$t = 4.52 \times 10^{24} / 1.5 \times 10^{19}$ C1

$= 3.0 \times 10^5$ s A1

25.

(a) **(i)** number of moles/amount of substance B1

(ii) kelvin temperature/absolute temperature/thermodynamic temperature B1

(b) $pV = nRT$

$4.9 \times 10^5 \times 2.4 \times 10^3 \times 10^{-6} = n \times 8.31 \times 373$ B1

$n = 0.38$ (mol) C1

number of molecules or $N = 0.38 \times 6.02 \times 10^{23} = 2.3 \times 10^{23}$ A1 *or*

$pV = NkT$ (C1)

$4.9 \times 10^5 \times 2.4 \times 10^3 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 373$ (M1)

number of molecules or $N = 2.3 \times 10^{23}$ (A1)

(c) volume occupied by one molecule $= (2.4 \times 10^3) / (2.3 \times 10^{23})$ C1

$= 1.04 \times 10^{-20}\ \text{cm}^3$

mean spacing $= (1.04 \times 10^{-20})^{1/3}$ C1

$= 2.2 \times 10^{-7}$ cm (*allow 1 s.f.*) A1

(*allow other dimensionally correct methods e.g.* $V = (4/3)\pi r^3$)

26.

4(a)	random/haphazard	**B1**
	constant velocity **or** speed in a straight line between collisions **or** distribution of speeds/different directions	**B1**
4(b)	(small) specks of light/bright specks/pollen grains/dust particles/smoke particles	**M1**
	moving haphazardly/randomly/jerky/in a zigzag fashion	**A1**
4(c)(i)	$pV = \frac{1}{3} Nm\langle c^2\rangle$ $1.05 \times 10^5 \times 0.0240 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2\rangle$	**C1**
	$\langle c^2\rangle = 1.89 \times 10^6$	**C1**
	or	
	$\frac{1}{2} m\langle c^2\rangle = (3/2)kT$ $0.5 \times (4.00 \times 10^{-3} / 6.02 \times 10^{23}) \times \langle c^2\rangle = 1.5 \times 1.38 \times 10^{-23} \times 300$	**(C1)**
	$\langle c^2\rangle = 1.87 \times 10^6$	**(C1)**
	or	
	$nRT = \frac{1}{3} Nm\langle c^2\rangle$ $1.00 \times 8.31 \times 300 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2\rangle$	**(C1)**
	$\langle c^2\rangle = 1.87 \times 10^6$	**(C1)**
	$c_{r.m.s.} = 1.37 \times 10^3\ \text{m s}^{-1}$	**A1**
4(c)(ii)	$\langle c^2\rangle \propto T$	**C1**
	$\langle c^2\rangle$ at 177 °C $= 1.89 \times 10^6 \times (450 / 300)$	**C1**
	$c_{r.m.s.}$ at 177 °C $= 1.68 \times 10^3\ \text{m s}^{-1}$	**A1**

27.

2(a)(i)	mean/average square speed/velocity	B1
2(a)(ii)	$pV = NkT$ **or** $pV = nRT$	B1
	$\rho = Nm / V$ **or** $\rho = nN_Am / V$ **and** $k = nR / N$	B1
	$E_K = ½ m\langle c^2 \rangle$ with algebra to $(3 / 2)kT$	B1
2(b)(i)	no (external) work done **or** $\Delta U = q$ **or** $w = 0$	B1
	$q = N_A \times (3 / 2)k \times 1.0$	M1
	$N_Ak = R$ so $q = (3 / 2)R$	A1
2(b)(ii)	specific heat capacity $= \{(3 / 2) \times R\} / 0.028$	C1
	$= 450$ J kg^{-1} K^{-1}	A1

28.

2(a)	$pV = nRT$ $T = (5.60 \times 10^5 \times 3.80 \times 10^{-2}) / (5.12 \times 8.31)$	C1
	$T =$ **500 K**	A1
2(b)(i)	V / T is constant $V = (3.80 \times 10^4) \times (500 + 125) / 500$	C1
	$V =$ **4.75 × 10⁴** cm^3	A1
2(b)(ii)	(for ideal gas,) change in internal energy is change in (total) kinetic energy (of molecules)	B1
	$\Delta U = 3 / 2 \times 1.38 \times 10^{-23} \times 125 \times 5.12 \times 6.02 \times 10^{23}$	C1
	= **7980** J	A1
2(c)(i)	$w = p\Delta V$ $= 5.60 \times 10^5 \times (4.75 - 3.80) \times 10^{-2}$	C1
	= **5320** J	A1
2(c)(ii)	total = 7980 + 5320 = **13300** J	A1

29.

2(a)	no intermolecular forces (so no potential energy)	B1
2(b)(i)	mean square speed (of molecule(s))	B1
2(b)(ii)	kelvin/thermodynamic/absolute <u>temperature</u>	B1
2(c)(i)1.	$pV = NkT$	C1
	$4.7 \times 10^{-2} \times 2.6 \times 10^5 = N \times 1.38 \times 10^{-23} \times 446$	C1
	or	
	$pV = nRT$ **and** $N = nN_A$ $4.7 \times 10^{-2} \times 2.6 \times 10^5 = n \times 8.31 \times 446$ $n = 3.3$ (mol)	(C1)
	$N = 3.3 \times 6.02 \times 10^{23}$	(C1)
	$N = 2.0 \times 10^{24}$	A1
2(c)(i)2.	average increase $= 2900 / (2.0 \times 10^{24})$ $= 1.5 \times 10^{-21}$ J	A1

2(c)(ii)	$\Delta E_K = (3/2)k\,(\Delta)T$ $1.5 \times 10^{-21} = (3/2) \times 1.38 \times 10^{-23} \times (\Delta)T$	C1
	$(\Delta)T$ in range 70–72 K	C1
	$T = 173 + 273 + 70$ $= 520$ K	A1

30.

2(a)	gas that obeys equation pV = constant × T	M1
	symbols p, V and T explained	A1
2(b)(i)	$pV = ⅓\, Nm\langle c^2\rangle$ **and** $M = Nm$ (and so) $p = ⅓\rho\langle c^2\rangle$	C1
	$2.12 \times 10^7 = ⅓ \times [3.20/(1.84 \times 10^{-2})] \times \langle c^2\rangle$	C1
	$c_{r.m.s.} = 605\,\text{m}\,\text{s}^{-1}$	A1
2(b)(ii)	**1.** $pV = nRT$ **and** $T = (22 + 273)$ K	C1
	$n = (2.12 \times 10^7 \times 1.84 \times 10^{-2})/(8.31 \times 295)$ $= 159$ mol	A1
	2. mass $= 3.20/(159 \times 6.02 \times 10^{23})$ **or** mass $= [2 \times (3/2) \times 1.38 \times 10^{-23} \times 295]/605^2$	C1
	mass $= 3.34 \times 10^{-26}$ kg	A1
2(c)	$A = (3.34 \times 10^{-26})/(1.66 \times 10^{-27})$ $= 20$	A1

31.

2(a)(i)	gas obeys formula pV/T = constant	M1
	symbols V and T explained	A1
2(a)(ii)	mean-square-speed (of atoms / molecules)	B1
2(b)(i)	use of $T = 393$	C1
	$pV = nRT$	C1
	$2.4 \times 10^5 \times 6.8 \times 10^{-3} = n \times 8.31 \times 393$ and $N = n \times 6.02 \times 10^{23} = 3.0 \times 10^{23}$	A1
	or $pV = NkT$	(C1)
	$2.4 \times 10^5 \times 6.8 \times 10^{-3} = N \times 1.38 \text{ x } 10^{-23} \times 393$ hence $N = 3.0 \times 10^{23}$	(A1)
2(b)(ii)	volume of one atom $= 4/3\pi r^3$	C1
	volume occupied $= 3.0 \times 10^{23} \times 4/3 \times \pi \times (3.2 \times 10^{-11})^3$ $= 4 \times 10^{-8}\,\text{m}^3$	A1
2(b)(iii)	assumption: volume of atoms negligible compared to volume of container / cylinder	B1
	$4 \times 10^{-8}\,(\text{m}^3) << 6.8 \times 10^{-3}\,(\text{m}^3)$ so yes	B1

32.

2(a)(i)	$pV = nRT$	**C1**
	$n = (3.0 \times 10^5 \times 210 \times 10^{-6}) / (8.31 \times 270)$	**C1**
	$= 0.028$ mol	**A1**
2(a)(ii)	$V \propto T$ **or** $T = pV / nR$ with value of n from **(i)**	**C1**
	$T = (140 / 210) \times 270$ **or** $T = (3.0 \times 10^5 \times 140 \times 10^{-6}) / (8.31 \times 0.028)$ $= 180$ K	**A1**
2(a)(iii)	$W = p\Delta V$ $= 3.0 \times 10^5 \times (210 - 140) \times 10^{-6}$	**C1**
	$= 21$ J	**A1**
2(b)	$\Delta U = w + q$	**C1**
	$= 21 - 53$	**C1**
	or	
	$\Delta U = (nN_A) \times (3/2)k\Delta T$	**(C1)**
	$= (0.0281 \times 6.02 \times 10^{23}) \times (3 / 2) \times 1.38 \times 10^{-23} \times (180 - 270)$	**(C1)**
	or	
	$\Delta U = (3/2)nR\Delta T$	**(C1)**
	$= (3/2) \times 0.0281 \times 8.31 \times (180 - 270)$	**(C1)**
	$\Delta U = (-)32$ J	**A1**

33.

2(a)	(total volume of molecules is) negligible	**M1**
	compared with volume occupied by the gas	**A1**
2(b)(i)	$pV = NkT$	**C1**
	$4.60 \times 10^5 \times 2.40 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times (273 + 23)$	**C1**
	or	
	$pV = nRT$	**(C1)**
	$4.60 \times 10^5 \times 2.40 \times 10^{-2} = n \times 8.31 \times (273 + 23)$ $n = 4.49$ (mol) $N = nN_A$ $= 4.49 \times 6.02 \times 10^{23}$	**(C1)**
	$N = 2.7 \times 10^{24}$	**A1**

2(b)(ii)	volume of one atom $= d^3$ ($= 2.7 \times 10^{-29}$ m^3)	**C1**
	volume of all atoms $= 2.7 \times 10^{-29} \times 2.7 \times 10^{24}$	**C1**
	$= 7 \times 10^{-5}$ m^3	**A1**
	or	
	volume of one atom $= (4/3)\pi r^3$ ($= 1.41 \times 10^{-29}$ m^3)	**(C1)**
	volume of all atoms $= 2.7 \times 10^{24} \times 1.41 \times 10^{-29}$	**(C1)**
	$= 4 \times 10^{-5}$ m^3	**(A1)**
2(c)	numerical comparison between answer to **(b)(ii)** and 2.4×10^{-2} (m^3) showing **(b)(ii)** is <u>much</u> less than 2.4×10^{-2} (m^3)	**B1**

34.

2(a)(i)	specks of light moving haphazardly	**B1**
2(a)(ii)	(gas) molecules collide with (smoke) particles **or** random motion of the (gas) molecules	**M1**
	<u>causes</u> the (haphazard) motion of the smoke particles **or** <u>causes</u> the smoke particles to change direction	**A1**
2(b)(i)	$pV = nRT$	**C1**
	$n = (3.51 \times 10^5 \times 2.40 \times 10^{-3}) / (8.31 \times 290)$ **or** $n = (3.75 \times 10^5 \times 2.40 \times 10^{-3}) / (8.31 \times 310)$	**C1**
	or	
	$pV = NkT$	**(C1)**
	$n = (3.51 \times 10^5 \times 2.40 \times 10^{-3}) / (1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 290)$ **or** $n = (3.75 \times 10^5 \times 2.40 \times 10^{-3}) / (1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 310)$	**(C1)**
	$n = 0.350$ mol **or** 0.349 mol	**A1**
2(b)(ii)	energy transfer = (0.349 **or** 0.35) $\times 12.5 \times (310 - 290)$	**C1**
	= 87.3 J **or** 87.5 J	**A1**
2(c)(i)	zero	**A1**
2(c)(ii)	87.3 J **or** 87.5 J	**A1**
	increase	**B1**

35.

2(a)	$n = 110 / 0.032$ or $110000 / 32$ or 3440	**C1**
	$pV = nRT$	**C1**
	$T = (1.0 \times 10^5 \times 85) / (8.31 \times (110 / 0.032)) = 300$ K	**A1**
2(b)	$E = mc\Delta\theta$ $= 110 \times 0.66 \times 50$	**C1**
	= **3600** J	**A1**
2(c)	Any 3 from: • molecule collides with wall • momentum of molecule changes during collision (with wall) • force on molecule so force on wall • many forces act over surface area of container exerting a pressure	**B3**

2(d)	$KE \propto T$ $v \propto \sqrt{T}$	C1
	ratio $= \sqrt{(350 / 300)}$ $=$ **1.1**	A1

36.

2(a)	ρ: Nm / V	B1
	⅓: molecules move in three dimensions (not one) so ⅓ in any (one) direction	B1
	$<c^2>$: molecules have different speeds so take average	M1
	of (speed)^2	A1
2(b)	$pV = NkT$	C1
	$N = (3.0 \times 10^5 \times 6.0 \times 10^{-3}) / (1.38 \times 10^{-23} \times 290)$	C1
	$= 4.5 \times 10^{23}$	C1
	mass $= 20.7 / (4.5 \times 10^{23})$ $= 4.6 \times 10^{-23}$ g	A1

37.

2(a)	sum of potential energy and kinetic energy (of particles)	B1
	(total) energy of random motion of particles	B1
2(b)(i)	$pV = nRT$	C1
	$2.60 \times 10^5 \times 2.30 \times 10^{-3} = n \times 8.31 \times 180$ $n = 0.400$ mol	A1
2(b)(ii)	$(2.30 \times 10^{-3}) / 180 = (3.80 \times 10^{-3}) / T$ **or** $2.60 \times 10^5 \times 3.80 \times 10^{-3} = 0.400 \times 8.31 \times T$	C1
	$T = 297$ K	A1
2(c)(i)	$\Delta W = p\Delta V$ $= 2.60 \times 10^5 \times (2.30 - 3.80) \times 10^{-3}$	C1
	$= (-)390$ J	A1
	negative because work is done by gas **or** negative because work is done against atmospheric pressure **or** negative because volume of gas increases	B1
2(c)(ii)	$\Delta U = (980 - 390)$ $= 590$ J	A1

38.

2(a)(i)	$pV = NkT$ or $pV = nRT$ <u>and</u> $N = nN_A$ $N = \dfrac{2.3 \times 10^5 \times 3.5 \times 10^{-3}}{1.38 \times 10^{-23} \times 294}$	C1
	= 2.0 × 10^{23}	A1
2(a)(ii)	$pV = \dfrac{1}{3}Nmc^2$ $c^2 = \dfrac{3 \times 2.3 \times 10^5 \times 3.5 \times 10^{-3}}{2.0 \times 10^{23} \times 40 \times 1.66 \times 10^{-27}}$ = 182 000 r.m.s. speed = **430** m s^{-1}	C1
	or $\frac{1}{2}mc^2 = \frac{3}{2}kT$	A1
	$c^2 = \dfrac{3 \times 1.38 \times 10^{-23} \times 294}{40 \times 1.66 \times 10^{-27}}$ = 183 000	(C1)
	r.m.s.speed = 430 m s^{-1}	(A1)
2(b)	$c^2 = \dfrac{3 \times 2.0 \times 10^{23} \times 1.38 \times 10^{-23} \times (294 + 84)}{2.0 \times 10^{23} \times 40 \times 1.66 \times 10^{-27}}$ $c^2 = 236\,000$ $c = 485$	C1
	$ratio\left(= \dfrac{485}{430}\right) = 1.1$	A1
	OR $v \propto \sqrt{T}$ or $v^2 \propto T$	(C1)
	$ratio = \sqrt{\dfrac{273 + 21 + 84}{273 + 21}}$ or $\sqrt{\dfrac{378}{294}}$ ratio = **1.1**	(A1)

39.

2(a)	$pV = NkT$	C1
	$N = (1.8 \times 10^{-3} \times 3.3 \times 10^5) / (1.38 \times 10^{-23} \times 310) = 1.4 \times 10^{23}$	A1
	or	
	$pV = nRT$ **and** $nN_A = N$	(C1)
	$N = (1.8 \times 10^{-3} \times 3.3 \times 10^5 \times 6.02 \times 10^{23}) / (8.31 \times 310) = 1.4 \times 10^{23}$	(A1)
2(b)	speed of molecule decreases on impact with moving piston	B1
	mean square speed (directly) proportional to (thermodynamic) temperature **or** mean square speed (directly) proportional to kinetic energy (of molecules) **or** kinetic energy (of molecules) (directly) proportional to (thermodynamic) temperature	B1
	kinetic energy (of molecules) decreases (so temperature decreases)	B1

2(c)(i)	$\Delta U = 3/2 \times k \times \Delta T \times N$	C1
	$= 3/2 \times 1.38 \times 10^{-23} \times (288 - 310) \times 1.4 \times 10^{23}$	C1
	$= -64$ J	A1
2(c)(ii)	decrease in internal energy is less than work done by gas	M1
	(thermal energy is) transferred <u>to</u> the gas (during the expansion)	A1

40.

2(a)	$pV = nRT$	C1
	$pV = nRT$ **and** $N = nN_A$ **or** $pV = NkT$	C1
	$3.1 \times 10^{-3} \times 8.5 \times 10^5 = (N \times 290 \times 8.31) / (6.02 \times 10^{23})$ so $N = 6.6 \times 10^{23}$ **or** $3.1 \times 10^{-3} \times 8.5 \times 10^5 = N \times 1.38 \times 10^{-23} \times 290$ so $N = 6.6 \times 10^{23}$	A1
2(b)(i)	$(3.1 \times 10^{-3} \times 8.5 \times 10^5) / 290 = (6.3 \times 10^{-3} \times 2.7 \times 10^5) / T$ so $T = 190$ K **or** $6.3 \times 10^{-3} \times 2.7 \times 10^5 = 6.6 \times 10^{23} \times 1.38 \times 10^{-23} \times T$ so $T = 190$ K	A1
2(b)(ii)	$\Delta U = 3/2 \times k \times \Delta T \times N$	C1
	$= 3/2 \times 1.38 \times 10^{-23} \times (190 - 290) \times 6.6 \times 10^{23}$	C1
	$= -1400$ J	A1
2(c)	$\Delta U = q + w$	M1
	$q = 0$ so $\Delta U = w$	A1

41.

3(a)(i)	no loss of <u>kinetic</u> energy	B1
3(a)(ii)	• <u>molecules</u> have negligible volume (compared with gas/container) • no forces between <u>molecules</u> (except during collisions) • <u>molecules</u> are in random motion • collisions are instantaneous *Any two points, 1 mark each*	B2
3(b)(i)	$2mu$	A1
3(b)(ii)	$2L / u$	A1
3(b)(iii)	force = change in momentum / time = $2mu / (2L / u)$ $= mu^2 / L$	A1
3(b)(iv)	pressure = force / area = $(mu^2 / L) / L^2$ $= mu^2 / L^3$	A1
3(c)	$pV = NkT$	C1
	$NkT = \frac{1}{3}Nm\langle c^2\rangle$ leading to $\frac{1}{2}m\langle c^2\rangle = (3/2)kT$ <u>and</u> $\frac{1}{2}m\langle c^2\rangle = E_K$	A1
3(d)	$\frac{1}{2} \times 3.34 \times 10^{-27} \times \langle c^2\rangle = (3/2) \times 1.38 \times 10^{-23} \times (25 + 273)$	C1
	r.m.s. speed $= 1.9 \times 10^3$ m s^{-1}	A1

42.

3(a)	p = pressure (of gas), V = volume (of gas) **and** k = Boltzmann constant	**B1**
	N = number of molecules	**B1**
	T = thermodynamic temperature	**B1**
3(b)	($pV = NkT$ and $pV = ⅓Nm\langle c^2\rangle$ leading to) $NkT = ⅓Nm\langle c^2\rangle$	**M1**
	algebra leading to $(3/2)kT = ½m\langle c^2\rangle$ **and** use of $½m\langle c^2\rangle = E_K$ leading to $(3/2)kT = E_K$	**A1**
3(c)(i)	T = 296 K	**C1**
	$½m\langle c^2\rangle = (3/2)kT$ $½ \times 5.31 \times 10^{-26} \times u^2 = (3/2) \times 1.38 \times 10^{-23} \times 296$	**C1**
	u = 480 m s^{-1}	**A1**
3(c)(ii)	line passing through (P, u)	**B1**
	horizontal straight line	**B1**

43.

2(a)(i)	(gas that obeys) $pV \propto T$ (for all values of p, V and T)	**M1**
	where T is thermodynamic temperature	**A1**
2(a)(ii)	temperature = –273.15 °C	**A1**
2(b)(i)	$pV = NkT$	**C1**
	$N = (1.37 \times 10^5 \times 0.640) / (1.38 \times 10^{-23} \times (227 + 273))$	**C1**
	$= 1.27 \times 10^{25}$	**A1**
2(b)(ii)	mass $= 0.0424 / (1.27 \times 10^{25})$ $= 3.34 \times 10^{-27}$ kg	**A1**
2(b)(iii)	$½m\langle c^2\rangle = (3/2)kT$	**C1**
	$3.34 \times 10^{-27} \times v^2 = 3 \times 1.38 \times 10^{-23} \times 500$	**C1**
	v = 2490 m s^{-1}	**A1**
	or	
	$pV = ⅓(Nm)\langle c^2\rangle$ <u>and</u> Nm = mass of gas	**(C1)**
	$0.0424 \times v^2 = 3 \times 1.37 \times 10^5 \times 0.640$	**(C1)**
	v = 2490 m s^{-1}	**(A1)**
2(c)	sketch: line from (0, 0) to (500, v)	**B1**
	line with decreasing positive gradient throughout	**B1**